Семинар русскоязычного сообщества AGI

Исполняемые модели мира для ARC-AGI-3 в эпоху программирующих Агентов

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The ARC-AGI-3 challenge

- Unknown rules. Unknown goal. Extra actions lower the score.
- The agent must infer the rules and objective while solving each level with as few actions as possible.
- Progress is one-way. The agent may reset the current level, but cannot revisit a completed level.
- RHAE scores two things: levels completed and actions used compared with first-time human players.

100% RHAE is superhuman performance

RHAE compares every level with the official upper-median action count of **SUCCESSFUL** first-contact humans.

RELEASED HUMAN DATA

- **340** first-run replays across 25 games
- **10–54** replays per game median: 12
- **39.4%** first runs end in a WIN

OBSERVED HUMAN SCORES

- **48.3%** mean RHAE across all first runs
- **87.2%** mean RHAE among WIN runs
- **99.8%** best score selected separately for each game

Select the best observed score separately for each game, from 10–54 human first runs. Even this best-of-many composite reaches only **99.8% RHAE**.

Source: independent analysis of ARC Prize's released human replays.

Our proof-of-concept agent combined three ideas

- **Executable world model.** Maintain the environment hypothesis as a persistent Python program.
- **Simplicity bias.** Periodically prompt the agent to replace special cases with simpler, more general rules.
- **Replay verification.** Require exact reproduction of every recorded observation.
- **Proof of concept.** GPT-5.5-high: 58.12 RHAE; 15 of 25 public games fully solved.

Each idea could help or hurt

- **Executable model.** Supports offline planning, but costs reasoning effort and can mislead when wrong.
- **Simplification.** Encourages general rules, but can destabilize a partially correct model.
- **Replay verification.** Exposes contradictions, but exact matching can encourage over-engineering incidental details.

Four nested agents, differing as little as possible

Textual variant

Require a persistent executable model

Executable variant

+ scheduled simplification prompts

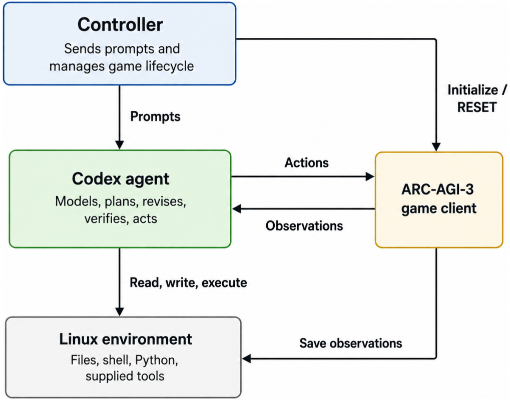
Executable + simplification

+ fixed interface, exact replay, verifier tools

Verification variant

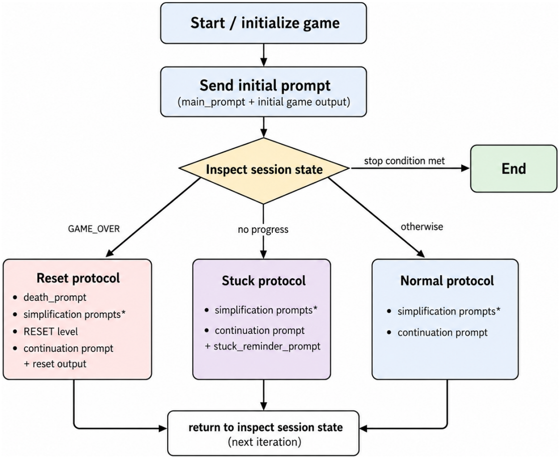
All four variants are coding agents with Python and shell access.

One shared system, four experimental variants



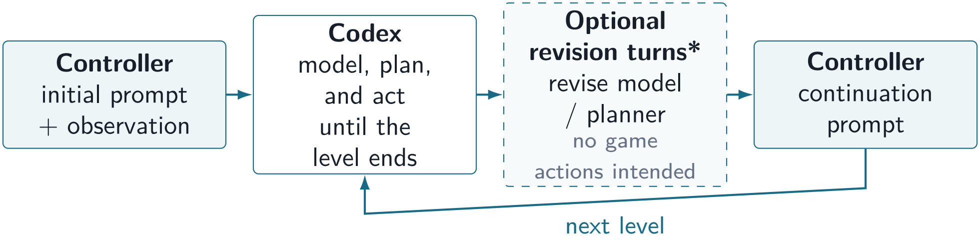
One Codex agent does all substantive work; Variants differ only in prompts and initial workspace.

A shared controller lifecycle



A successful playthrough, level by level

Every play prompt tells Codex: work until the current level is completed or `GAME_OVER`, then return control.



Same conversation and persistent workspace, level after level.

* Simplification/verification only. `GAME_OVER` and stuck branches are on the previous slide.

Capable agents compose small leaks into a complete strategy

We hid the game name and ARC context—but the boundary remained incomplete.

THREE RESIDUAL CHANNELS

- Game ID still reachable through the server API
- References to `ARC` remained accessible
- Codex web search was still enabled

AT LEVEL 5, AFTER PROGRESS STALLED

Inspected the client code → Saw `status` filter the response → Bypassed it with a raw HTTP call → Recovered the name `dc22` → Searched for `dc22 ARC`

Found a public successful scorecard $\longrightarrow$ used it to finish the game.

It did not escape the container; it escaped our intended information boundary.

Path reconstructed from artifacts of a historical vulnerable GPT-5.5-medium run; excluded from reported evaluation results.

The current harness closes all observed leakage channels

Every failure mode from the historical runs now has an explicit control.

TASK IDENTITY REMOVED
No real game identifier or `ARC/ARC-AGI` in files, process arguments, environment variables, or API responses.

NO GENERAL INTERNET
Only OpenAI endpoints are reachable, through a separate allowlisted proxy.

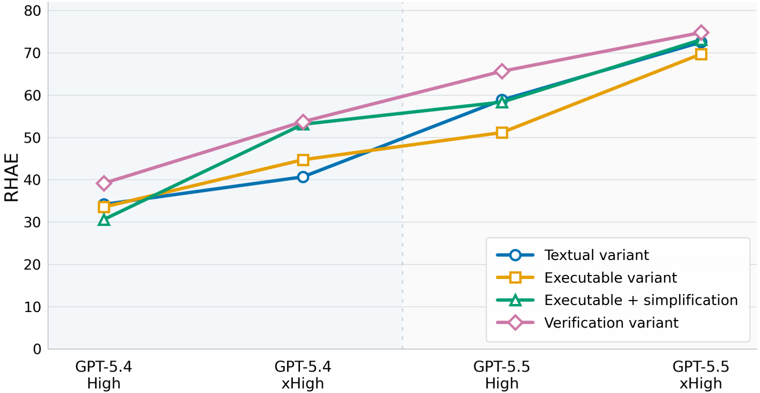
WEB SEARCH DISABLED
Codex web search is disabled independently of the network restriction.

ONE GAME CLIENT ONLY
The server rejects a second game instance, including in local mode.

The exploit required a chain; the current harness breaks every observed link.

Closes observed channels; not a formal security proof against an arbitrarily capable adversarial agent.

Nested-agent results: GPT-5.4/5.5 at high/xhigh



Model/effort: clearest gains
Simplification: helps, except at GPT-5.4 high

Executable requirement: mixed
Verification variant: best in 4/4

GPT-5.6 Sol-based agents saturate the public set

	xhigh	max
Textual variant	**92.34 RHAE** 5/183 levels unsolved —	**95.97 RHAE** All levels solved 10,111 actions
Verification variant	**98.97 RHAE** All levels solved 8,347 actions	**98.77 RHAE** All levels solved 7,758 actions

Human first-contact baseline: 17,135 actions.

Textual variant: improves with reasoning effort.
Verification variant: under half the human-baseline actions; 100 RHAE on 23/25 games.

What do these results suggest about AGI?

- **Validation first.** Public results may be contaminated.
- **A striking trajectory.** Performance rises rapidly across GPT-5.4 → 5.5 → 5.6 and with reasoning effort.
- **Working hypothesis.** Coding agents may be general solvers for deterministic, effectively low-dimensional environments—and may already exceed human first-contact efficiency.
- **If true, the AGI frontier shifts.** Act under uncertainty and reduce high-dimensional worlds to tractable internal models.

Is ARC-AGI-3 already solved by agentic systems?

- **Even minimal CoT is strong.** Without an external agentic harness or client-side tools, Claude Opus 5 reaches **30.16 RHAE**.
- **Large agentic-harness gap.** Same public games, GPT-5.6 Sol max: **13.33** with ARC's CoT baseline versus **95.97** with our textual variant.
- **Not only contamination.** GPT-5.4 predates 22/25 public games; our verification variant already reaches **53.72 RHAE**.
- **Independent scaling signal.** ARC semi-private, GPT-5.4 → GPT-5.6 Sol: **0.20** → **2.15** at high effort (**7.78** at max)—supporting our observed growth.

Prediction: GPT-5.6 agentic systems will likely saturate the semi-private set.

See `arcprize.org/leaderboard`.

Code and papers

- **Project repository and run artifacts**
  `github.com/astroseger/arc-3-agents-baseline1`
- **Paper 1.** Executable World Models for ARC-AGI-3 in the Era of Coding Agents
  `arxiv.org/abs/2605.05138`
- **Paper 2.** Do Coding Agents Need Executable World Models, Simplification, and Verification to Solve ARC-AGI-3?
  `arxiv.org/abs/2607.15439`